Hands-on Lab – E490 Equipment Rack Access

Background / Scenario

This lab will guide you through the steps to access the networking equipment racks in the E490 lab.

Instructions

Part 1: Set Up the Cabling

Step 1:

a. You will be first needing a Cisco Console Cable as shown below. The Female end of the cable will be connecting to the back of the Lab PC serial port (Male).

b. As for the ethernet or RJ45 end, it will be connecting to either port that label D001, D002, etc.



Figure 1: CISCO Console Cable

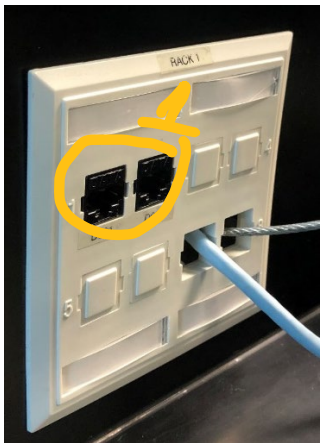


Figure 2: RJ45 ports

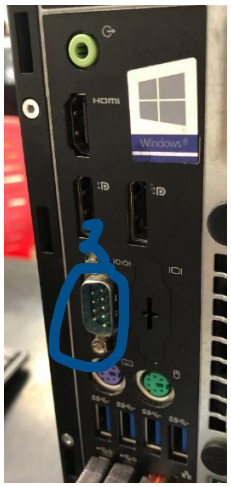


Figure 3: Back of the Lab PC

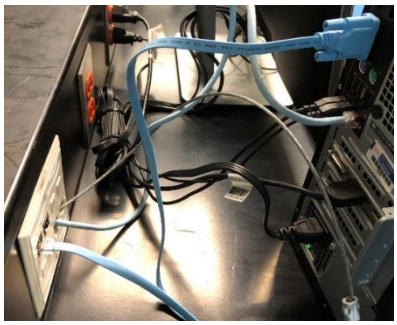


Figure 4: RJ45 Port and Lab PC Serial Port are connected using Cisco Serial Cable and either a Straight thru or Crossover cable depending upon the topology given.

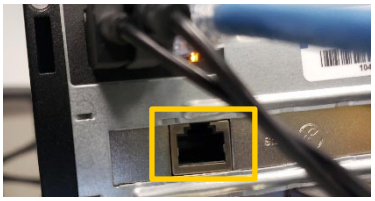


Figure 5: Second NIC card interface on the desktop. Could be the "Ethernet Network 4" interface on the Win11 virtual machine (VM)

Step 2:

a. Find your rack by checking the label on top of the RJ45 outlet that you connected to earlier. Rack # represents the rack that you will be working on. There are a total of 6 racks, which match the number of rows that the ENG 490 class has.

b. Figure 5 illustrates the rack after the front door of the rack cabinet is opened

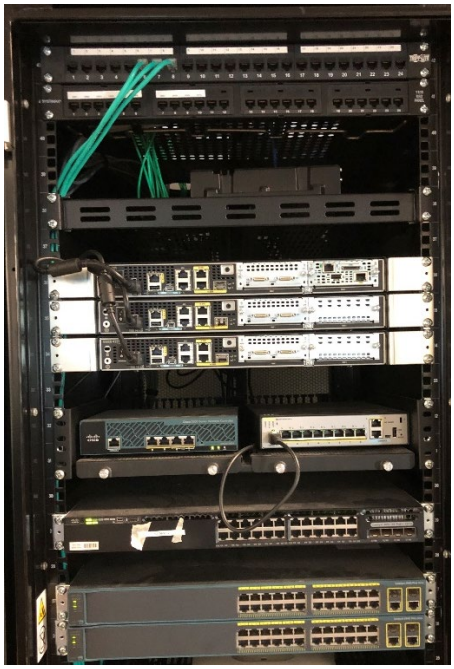


Figure 6: Cisco Rack



Figure 7: RJ45 Port Panels



Figure 8: Cisco 4321 Routers

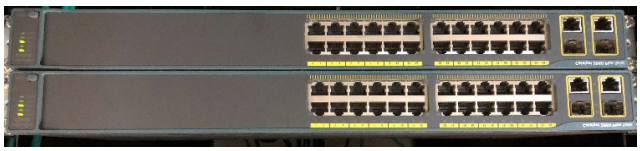


Figure 9: Cisco Catalyst 2960 Plus Series Switches

Step 3:

a. You will be using the green Straight-Through Cable, as shown in Figure 9, that is hanged on the far right of all the racks. From there, connect either end of the cable to the corresponding port on the Port Panel.

b. If you connected your Cisco Serial Cable to D001, then on the Port Panel, you will be connecting the green Straight-Through Cable to D001, as seen in Figure 10, on there as well.

c.



Figure 10: Green Straight-Through Cable



Figure 11: Port Panel up close

Step 4:

a. After one end of the green cable is connected with the Port Panel, you can now connect the other end with either the switch (Figure 9) and router (Figure 10) to have console access with them.

b. For the router, you must connect the specify port circle in Figure 12. If you connect to the port above it, it will not work because that port is for AUX only. For the switch, you need to connect to the back far left of the switch, as circle in Figure 13.

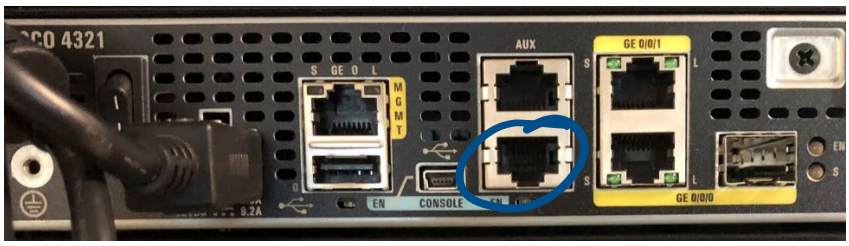


Figure 12: Cisco 4321 Router up close with direction



Figure 13: Back of the 2960 Switch up close with direction

Step 5:

a. After you finish connecting your other end to either the router or switch, it should be as follow:



Figure 14: Green Straight-Through Cable connecting from Port Panel D001 to Cisco 4321 Router console port



Figure 15: Green Straight-Through Cable connecting from Port Panel D001 to Cisco Catalyst 2960 Switch console port

Step 6:

a. Cable the activity as it appears in the topology below

b. NOTE: You will be connecting the PCs through the table jack to the patch panel on the equipment rack and from the patch panel to the switch or router Console and Ethernet ports.

Topology:

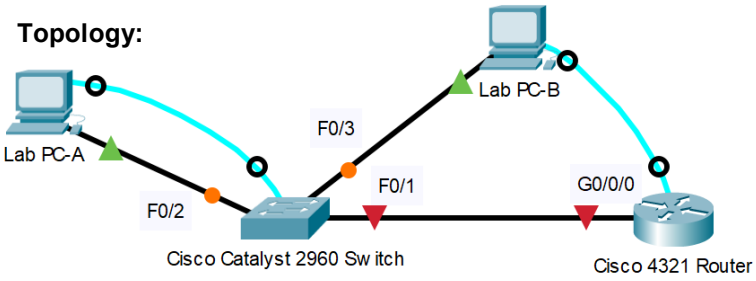
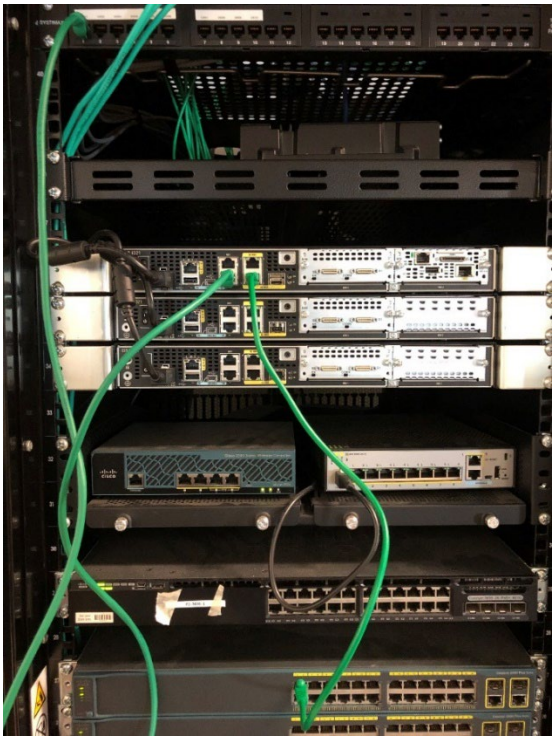


Figure 16: Connections between Switch F0/0 and Router G0/0 with Lab PCs console connecting to the Router and Switch.



Part 2: Configure Devices and Verify Connectivity

In Part 2, you will configure the router and the switch and verify connectivity. You need to use Putty on the desktop for this part of the lab.

Step 1: Configure the router

a. Console into the routers and enable privileged EXEC mode.

b. Enter configuration mode.

c. Assign a device name to the router.

d. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

e. Assign **class** as the privileged EXEC encrypted password.

f. Assign **cisco** as the console password and enable login.

g. Assign **cisco** as the VTY password and enable login.

h. Encrypt the plaintext passwords.

i. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

j. Configure and activate the Ethernet interface on the router. Use **192.168.65.1 /24** as the address.

k. **DO NOT SAVE** the running-config to the startup-config.

Step 2: Configure the Switch

In this step, you will configure the hostname, the VLAN 1 interface and default gateway.

a. Console into the switch and enable privileged EXEC mode.

b. Enter configuration mode.

c. Assign a device name to the switch.

d. Disable DNS lookup to prevent the switch from attempting to translate incorrectly entered commands as though they were host names.

e. Configure and activate the VLAN interfaces on the switches. **Use 192.168.65.2 /24** as the address.

f. Configure the default gateway for the switch.

g. **DO NOT SAVE** the running-config to the startup-config.

Step 3:

a. Configure the IPv4 addressing on the PCs

b. PC-A: use 192.168.65.3 /24 as the address

c. PC-B: use 192.168.65.4 /24 as the address

Step 4: Verify connectivity to the router and the switch

a. Ping the router and the switch from one of the PCs.

b. Take a screenshot (use the desktop Snipping Tool) of the ping results. Each of your team members should upload the screenshot as their submission on Canvas for this this assignment.